

User Manual

CineBandit — Adaptive Movie Recommendation System

Who is this for? This guide is written for non-technical users. No programming knowledge is needed. If you can browse a website, you can use CineBandit.

1. What is CineBandit?

CineBandit is a movie recommendation application that learns your preferences as you use it. Unlike a static list, it pays attention to which movies you click on and which you skip, and adjusts its suggestions in real time — the more you interact with it, the better the recommendations become.

Think of it like a knowledgeable friend who remembers every movie you liked or disliked and uses that memory to suggest what to watch next.

1.1 What can it do?

Feature	What it means for you
Personalised recommendations	See a list of movies chosen specifically for your taste
Adaptive learning	The more you rate, the smarter the suggestions get
Live monitoring dashboard	Watch the system learn in real time (for the curious)
Pipeline console	See the underlying data pipelines running (for advanced users)

2. Getting Started

CineBandit runs on your computer using Docker. If someone has set it up for you, skip to Section 3. If you need to start it yourself, follow the steps below.

2.1 Starting the application

Before you begin Make sure Docker Desktop is installed and running on your computer. You will see the Docker whale icon in your taskbar.

Step 1: Open a terminal — On Mac: press Cmd+Space and type 'Terminal'. On Windows: press Win+R and type 'cmd'.

Step 2: Navigate to the project folder — Type: `cd path/to/cinebandit` and press Enter.

Step 3: Start all services — Type: `docker compose up -d` and press Enter. Wait about 60 seconds.

Step 4: Open the app — Open your web browser and go to: `http://localhost:8501`

Tip If the page doesn't load immediately, wait 30 more seconds and try again. All 9 services need to start up.

3. Using the App

The CineBandit dashboard has three sections accessible from the left sidebar. The main section you will use daily is Recommendations.

3.1 The Recommendations screen

This is your home screen. It shows a list of movies the system thinks you will enjoy based on your history.

Getting recommendations

1. Make sure your User ID is set in the sidebar (your administrator will tell you your ID).
2. Choose how many recommendations you want using the 'Number of suggestions' slider.
3. Click the 'Get Recommendations' button.
4. A list of movies appears, each with a title, genre, and confidence score.

Giving feedback — the most important thing you can do

After watching a recommended movie, come back and let CineBandit know what you thought. This is how the system learns.

Button	What it means	When to use it
✓ Liked it	You enjoyed the movie	After watching something you'd recommend to a friend
✗ Not for me	It wasn't your taste	If the movie wasn't enjoyable to you
Skip	You haven't watched it yet	If you want to leave a movie for later

Why feedback matters Every time you click Liked or Not for me, the system updates its model within seconds. You should notice better suggestions within 5–10 ratings.

3.2 The Monitoring screen

This screen shows live graphs of how well the system is performing. You do not need to do anything here — it updates automatically.

Chart	What it shows
Click-Through Rate (CTR)	The percentage of recommended movies you have clicked on. Higher is better. Target: above 55%.
Response time	How fast the system responds to your requests. Should always be under 1 second.
Drift score	How much the viewing patterns have changed recently. A high score triggers automatic retraining.

Good to know All monitoring charts refresh automatically every 15 seconds. You do not need to reload the page.

3.3 The Pipeline Console screen

This screen is for advanced users who want to see what is happening behind the scenes. It shows the status of automated data pipelines.

Pipeline	What it does
Data ingestion	Loads and validates the movie ratings data daily.
Drift detection	Checks whether viewing patterns have shifted significantly (runs every hour).
Model retraining	Updates the recommendation model automatically when patterns change.

Note If you see a pipeline with a red status, it usually means it is still running. Wait a few minutes and refresh.

4. Common Questions

Why am I getting the same recommendations every time?

This usually means you haven't submitted any feedback yet. The system starts with a general set of recommendations and personalises them as you rate movies. Try rating 5–10 movies and you should see the suggestions change.

The app won't load. What do I do?

- Check that Docker Desktop is running (look for the whale icon in your taskbar).
- Open a terminal and type: `docker compose ps` — all services should show 'Up'.
- If any service is not running, type: `docker compose restart`
- Wait 60 seconds, then try again at <http://localhost:8501>

How do I stop the application?

Open a terminal, navigate to the CineBandit folder, and type: `docker compose down`. This safely shuts down all services. Your ratings and history are saved automatically.

Where is my data stored?

All your ratings and preferences are stored locally in the PostgreSQL database running on your computer inside Docker. No data is sent to any external server.

How long until recommendations improve?

You should notice improvement after 5–10 feedback ratings. The system updates instantly — there is no delay. After 50+ ratings the personalisation becomes noticeably accurate.

5. Quick Reference

Task	Where to find it	What to do
Get movie suggestions	Recommendations screen	Set User ID → Set count → Click 'Get Recommendations'
Rate a movie	Recommendations screen	Click Liked / Not for me next to the movie
Check system health	Monitoring screen	View CTR and response time charts
See pipeline status	Pipeline Console screen	Check that all pipelines show green
Start the app	Terminal	<code>docker compose up -d</code>
Stop the app	Terminal	<code>docker compose down</code>
Restart a broken service	Terminal	<code>docker compose restart</code>

6. Glossary

Term	Plain English explanation
Recommendation	A movie the system thinks you will enjoy based on your history.
CTR (Click-Through Rate)	The fraction of recommended movies you actually clicked on. A higher CTR means better recommendations.
Feedback	Telling the system whether you liked or disliked a movie. This is how it learns.
Model	The algorithm inside CineBandit that decides which movies to suggest.
Pipeline	An automated sequence of tasks that prepares data and updates the model.
Drift	When viewing patterns change significantly. The system detects this and updates itself.
Docker	Software that packages the application so it runs the same way on any computer.

For technical support, refer to the project repository: github.com/chirag3211/cinebandit